Living Systems as Decentralized Economies

BME 129C Capstone — Sage Clokey — Spring 2026, UC Santa Cruz

Abstract

The Living Age is the transition from centralized mechanical civilization to a decentralized living world — where order emerges naturally through voluntary cooperation, conscience, culture, and regenerative systems rather than coercion or centralized control.

There are two types of order: decentralized and centralized. Centralized order requires uniformity. Decentralized order requires ordered diversity. Life is decentralized, and life must die to become centralized — it must be reduced to parts, and in doing so it loses what makes it valuable. The diversity of life is not random difference. It is based on local knowledge of time and place that no central planner can predict. Distributed nodes can plan, but central planners cannot. The distributed knowledge is coordinated by prices — the ratio between the voluntary exchange of anything between nodes. Living order is organized by choice. State order is dictated by control.

Living things are not machines. The differences are a feature, not a bug. The differences are how adaptation is possible. It is trial and error directed by distributed knowledge — just how Hayek explains the market at the human-to-human level with trade. Eugenics is the same fatal conceit: you cannot centrally plan life. If there is distributed knowledge in living things, it is in the DNA — and bioinformatics is the analysis of that genetic data. Bioinformatics is the study of distributed knowledge in living systems.

This capstone provides quantitative evidence — drawn from network topology, single-cell transcriptomics, agent-based metabolic simulation, genome-scale flux balance analysis, cross-species gene transfer, and cancer genomics — that the distributed architecture of biology is not a constraint to be overcome but the fundamental reason living systems outperform every centralized alternative we can design or simulate.

The Evidence: What the Data Shows

Layer 1: No Master Node — Network Topology

The E. coli gene regulatory network has no central controller. The degree distribution follows a power law (alpha 2.0–2.5), matching the Barabasi-Albert model for decentralized network growth. Hubs exist — CRP regulates 43 genes — but no hub dominates. Betweenness centrality is distributed across many nodes, not concentrated in one.

The robustness test is decisive. Under targeted removal of the most connected nodes, biological networks survive the loss of 37% of nodes before fragmenting. The star graph — the architecture of central planning — collapses at 1.9%. A 19:1 ratio. The distributed network routes around damage. The centralized network has a single point of failure.

Feed-forward loops are over-represented by massive Z-scores against 1,000 randomized networks. These are not accidental structures. They are evolved coordination motifs — Hayekian price signals that propagate local information without waiting for central command.

The self-regulation analysis shows that biological networks actively resist centralization. When hubs gain connections, their betweenness decreases — the network routes around them. Hub erosion. The invisible hand in topology.

Layer 1b: Cells as Economic Agents — Single-Cell Economy

Same genome, different output. PBMC single-cell RNA sequencing shows 8 distinct cell types occupying separate regions of gene expression space. No master cell assigned roles. Each cell differentiated in response to local signals — cytokine gradients, cell-cell contact, stochastic expression. Division of labor emerged from individual cells reading local conditions and making local decisions.

Shannon entropy varies across cell types: specialists concentrate transcriptional resources on narrow programs; generalists express broadly. The variation is the signal — it proves comparative advantage.

The communication network has no gatekeeper. Betweenness Gini of 0.0 — every cell type signals directly to every other. Removing any single cell type leaves 70–

90% of communication edges intact. No single removal is catastrophic. The same ligand is received differently by different cell types — Menger's subjective value at the molecular level. The value of a signal depends on who receives it, not on the molecule itself.

Layer 2: The Distributed Economy Outperforms the Planner

Thirteen metabolic pathway agents share a metabolite pool with no central allocator. Production rates oscillate early — the system discovering prices — then converge to stable, unequal values. No planner needed. The invisible hand finds equilibrium through local feedback.

Under perturbation, distributed allocation retains 71% of GDP. Centralized allocation retains 53%. An 18-point advantage for the market under stress. The planner's fixed plan is wrong after the perturbation, and it has no mechanism to discover the new optimum. The distributed agents self-correct through local price signals.

FBA using the genome-scale iML1515 model (2,712 reactions, 1,877 metabolites, 1,516 genes) is the omniscient planner — a linear program with perfect stoichiometric knowledge. It achieves 70% accuracy on real Keio knockout data. The 30% failure rate is structural: knowledge encoded in allosteric feedback, protein folding, expression timing, and chaperone requirements exists only in the local state of each molecular agent. The LP cannot capture it. Even the omniscient planner must compute shadow prices to solve its optimization — proving Hayek's point that prices contain essential information even the planner needs.

Shadow prices shift dramatically with context. NADH is cheap on glucose, expensive on acetate. Oxygen is free aerobically, most valuable anaerobically. The same molecule has different marginal value depending on the environment. Menger's subjective value computed from real metabolic data.

The Price System of the Cell — and What Happens When It Breaks

Living systems have a real price system operating at three tiers. Intracellular metabolite ratios (ATP/ADP, NAD+/NADH, AMP/ATP) function as cost of capital — emerging from the cell's own activity, not set by any authority. Intercellular signals (cytokines, morphogens, growth factors, oxygen tension) function as market prices — carrying tissue-level information that cells read locally. mTOR integrates all prices simultaneously and makes the grow-or-conserve decision — the entrepreneur reading the full local market.

Cancer is specifically the destruction of this price system. Using TCGA PanCancerAtlas data (10,967 samples, 32 cancer types), the most mutated genes map to price system components: receptors (EGFR, ERBB2, NOTCH1), integrators (PIK3CA, PTEN, MTOR, STK11), decision makers (TP53, RB1, CDKN2A), and spatial prices (FAT1, NF2). The tissue-specific variation is the finding: TP53 is mutated in 96% of ovarian cancers but 1% of thyroid. PIK3CA hits 52% in uterine but 3% in ovarian. Different tissues break different price components because different tissues rely on different prices. The disease is not the variant. The disease is the system that stopped reading the signal. Cancer is the calculation problem at the cellular level.

Layer 3: Comparative Advantage Across the Tree of Life

Each organism specializes in capabilities others lack. Coral exports biomineralization. Spider exports silk. Bacteria export cellulose. Planaria exports regeneration. No organism does everything. Ricardian comparative advantage at the molecular level.

Trade cost correlates with evolutionary distance — same-kingdom pairs trade easily, cross-kingdom pairs face high friction. The biological gravity model. Voluntary exchange succeeds; forced exchange (full codon optimization across high barriers) destroys the local knowledge encoded in rare codons. Trade blocs emerge spontaneously through Louvain community detection — phylogenetically related organisms cluster because compatible partners naturally trade more, not because anyone designed the partnership.

The Throughline

Every layer answers the same question from a different angle: does biology operate like a centrally planned economy or a free market?

The network structure says market — no master node, self-regulation, feed-forward price signals. Single cells say market — specialization, voluntary exchange, subjective value, distributed robustness. Metabolic allocation says market — distributed beats centralized under perturbation, even against an omniscient planner. Cancer genomics says market — disease is the destruction of the cellular price system, not the presence of bad parts. Cross-species trade says market — voluntary exchange succeeds, forced exchange fails, trade blocs emerge spontaneously.

Living things are not machines. The function is in the connections, not the parts. The differences are how adaptation is possible. The distributed knowledge is coordinated by prices. And the data shows it at every scale.

Life is the original decentralized network. The self-assembling internet. The adaptive computer that built itself, runs itself, repairs itself, and has been doing so for four billion years. We did not invent this architecture. We finally recognized it. The question now is not how to program life. It is how to join the computation — how to lead life to lead itself, and in doing so, grow something that lasts.